The uniform uncertainty principle and compressed sensing

Terence Tao · 5 December 2008

https://terrytao.wordpress.com/2008/12/05/the-uniform-uncertainty-principle-and-compressed-sensing/

tao2typ · converted to Typst 2026-10-14 (UTC)

This week I am in Seville, Spain, for a conference in harmonic analysis and related topics. My talk is titled "the uniform uncertainty principle and compressed sensing". The content of this talk overlaps substantially with my Ostrowski lecture on the same topic; the slides I prepared for the Seville lecture can be found here.

[*Update*, Dec 6: Some people have asked about my other lecture given in Seville, on structure and randomness in the prime numbers. This lecture is largely equivalent to the one posted here.]

Comments (8)

 **Anonymous** · 5 December 2008

Did you check the proof of the Goldbach conjecture submitted to arxiv today?

 **Terence Tao** · 6 December 2008

I only looked at it briefly, skipping to the last section which seems to be the only part that actually has to do with primes and the Goldbach problem. From the arguments given there, it appears the author only proves a somewhat weaker statement than the Goldbach conjecture, namely that there *exists* an even number greater than four which is the sum of two primes.

 **AA** · 7 December 2008

I copied the word "somewhat" from that comment and pasted it to my short list of "staggering understatements of the year 2008".

 **Harald Hanche-Olsen** · 8 December 2008

I think that calling that a staggering understatement is itself an understatement of epic proportions. And recursively, what I myself just wrote is probably an understatement too.
Anyway, if anybody wonders about Terry's whereabouts now that the Seville conference is over, I can reveal that he is currently in Trondheim, Norway, where he has delivered this year's Onsager lecture (after receiving his very own Onsager medal of course).

 **Anonymous** · 8 December 2008

Terence have you heard from Alexander Grothendieck? Any plans to meet him, to get him working on math again? He's gone from maths for a long time now, what is he doing currently?

 **Peter** · 8 December 2008

I also found that reply amusing :-) The same guy has over 700 pages of papers on the arxiv, and from a glancing at a couple of them, they seem to be filled with nonsense (although with many mathematically sophisticated words). It makes me wonder what's going through the head of someone who would learn so many words and then arrange them in (seemingly) meaningless sentences.

 **Igor Carron** · 29 December 2008

Terry,
Your lecture on Compressed Sensing at the Onsager Lecture series is at:
http://multimedie.adm.ntnu.no/mediasite/Viewer/?peid=dffa743e1d7446f7b493ea3205ddd41d
it cannot be read on Chrome and Opera, but they are available in chunks on Youtube:
http://nuit-blanche.blogspot.com/2008/12/cs-terence-tao-video-presentation-of.html
Another item that you might find interesting is a list of different technologies that are implementing CS in hardware (the single pixel camera at Rice being the prime example)
http://igorcarron.googlepages.com/compressedsensinghardware
Cheers,
Igor.

 **Terence Tao** · 29 December 2008

Dear Igor,
Thanks for the links!